

Python Tuple Practice — Basic to Hard

50 Python Tuple Practice Questions

Arranged from Basic → Hard • Each question includes example input and expected output.

Level	Difficulty	Focus	Questions
1	Basic	Tuple creation, indexing, slicing, counting, membership	10
2	Basic → Intermediate	Packing/unpacking, nested tuples, searching and transformations	10
3	Intermediate	Sorting, grouping, tuple-based records and multi-step logic	10
4	Advanced	Nested data, custom ordering, aggregation and algorithmic problems	8
5	Hard	Complex tuple structures and optimization-style problems	12

Level 1 — Basic

Tuple creation, indexing, slicing, counting, membership.

Q No.	Question	Example Input	Example Output
1	Create a tuple of integers.	Input: 1, 2, 3, 4	Output: (1, 2, 3, 4)
2	Find the length of a tuple.	Input: T=(10,20,30,40)	Output: 4
3	Access an element by index.	Input: T=('a','b','c'), index=1	Output: 'b'
4	Access the last element using negative indexing.	Input: T=(5,10,15,20)	Output: 20
5	Check whether a value exists in a tuple.	Input: T=(1,3,5,7), x=5	Output: True
6	Count occurrences of an element.	Input: T=(1,2,2,3,2), x=2	Output: 3
7	Find the first index of an element.	Input: T=('a','b','c','b'), x='b'	Output: 1
8	Slice a tuple.	Input: T=(0,1,2,3,4), T[1:4]	Output: (1,2,3)
9	Concatenate two tuples.	Input: A=(1,2), B=(3,4)	Output: (1,2,3,4)
10	Repeat a tuple three times.	Input: T=('x',)	Output: ('x','x','x')

Level 2 — Basic → Intermediate

Packing/unpacking, nested tuples, searching and transformations.

Q No.	Question	Example Input	Example Output
1	Unpack a tuple into three variables.	Input: T=(10,20,30)	Output: a=10, b=20, c=30
2	Swap two variables using tuple unpacking.	Input: a=5, b=9	Output: a=9, b=5
3	Convert a list to a tuple.	Input: [1,2,3]	Output: (1,2,3)
4	Convert a tuple to a list.	Input: (4,5,6)	Output: [4,5,6]
5	Find minimum and maximum values.	Input: T=(8,3,10,2)	Output: Min=2, Max=10
6	Find the sum of numeric tuple elements.	Input: T=(2,4,6,8)	Output: 20
7	Reverse a tuple.	Input: T=(1,2,3,4)	Output: (4,3,2,1)
8	Check whether a tuple is a palindrome.	Input: T=(1,2,3,2,1)	Output: True
9	Find all even values in a tuple.	Input: T=(1,2,4,7,8)	Output: (2,4,8)
10	Join a tuple of strings with spaces.	Input: ('Python','is','easy')	Output: 'Python is easy'

Level 3 — Intermediate

Sorting, grouping, tuple-based records and multi-step logic.

Q No.	Question	Example Input	Example Output
1	Sort a tuple of numbers.	Input: T=(5,1,4,2,3)	Output: (1,2,3,4,5)
2	Find the second largest distinct value.	Input: T=(10,5,8,10,3)	Output: 8
3	Remove duplicate values while returning a tuple.	Input: T=(1,2,2,3,1,4)	Output: (1,2,3,4)
4	Find common elements of two tuples.	Input: A=(1,2,3), B=(2,3,4)	Output: (2,3)
5	Find elements present only in the first tuple.	Input: A=(1,2,3), B=(2,4)	Output: (1,3)
6	Count how many tuples contain a given value.	Input: T=((1,2),(2,3),(4,5)), x=2	Output: 2
7	Flatten a one-level nested tuple.	Input: T=((1,2),(3,4),(5,6))	Output: (1,2,3,4,5,6)
8	Find the tuple with the largest sum.	Input: T=((1,2),(5,1),(2,2))	Output: (5,1)
9	Sort tuples by their second element.	Input: [(1,5),(2,2),(3,4)]	Output: [(2,2),(3,4),(1,5)]
10	Merge two tuples and remove duplicate values.	Input: A=(1,2,3), B=(3,4,2)	Output: (1,2,3,4)

Level 4 — Advanced

Nested data, custom ordering, aggregation and algorithmic problems.

Q No.	Question	Example Input	Example Output
1	Sort a list of tuples by multiple keys: first ascending, second descending.	Input: [(1,2),(1,5),(2,3),(2,1)]	Output: [(1,5),(1,2),(2,3),(2,1)]
2	Find the longest tuple in a list of tuples.	Input: [(1,2),(3,4,5),(6,)]	Output: (3,4,5)
3	Group tuples by their first element.	Input: [(1,'a'),(2,'b'),(1,'c')]	Output: {1:[('a'),('c')], 2:['b']}
4	Find pairs of tuple elements whose sum equals a target.	Input: T=(2,4,5,7,9), target=11	Output: (2,9), (4,7)
5	Find the tuple with the maximum second value.	Input: T=((1,10),(2,7),(3,15))	Output: (3,15)
6	Transpose a tuple of tuples representing a matrix.	Input: ((1,2,3),(4,5,6))	Output: ((1,4),(2,5),(3,6))
7	Calculate column-wise sums of a tuple matrix.	Input: ((1,2),(3,4),(5,6))	Output: (9,12)
8	Find duplicate tuples in a list.	Input: [(1,2),(3,4),(1,2),(5,6),(3,4)]	Output: {(1,2),(3,4)}

Level 5 — Hard

Complex tuple structures and optimization-style problems.

Q No.	Question	Example Input	Example Output
1	Find the k-th largest distinct value across nested tuples.	Input: ((1,9),(4,7),(9,2)), k=2	Output: 7
2	Find all pairs of tuples whose combined sum equals a target.	Input: [(1,2),(2,3),(3,4)], target=10	Output: ((1,2),(3,4))
3	Find the longest increasing subsequence of tuple values.	Input: T=(3,1,2,5,4,6)	Output: (1,2,5,6)
4	Find the median of numeric values stored in nested tuples.	Input: ((1,5),(2,9),(3,7))	Output: 6
5	Find the most frequent tuple in a list.	Input: [(1,2),(2,3),(1,2),(1,2)]	Output: (1,2)
6	Find the k smallest distinct values in a tuple.	Input: T=(8,3,5,3,1,9), k=3	Output: (1,3,5)
7	Find all nested tuples whose elements are all even.	Input: T=((2,4),(1,6),(8,10))	Output: ((2,4),(8,10))
8	Rotate a tuple to the left by k positions.	Input: T=(1,2,3,4,5), k=2	Output: (3,4,5,1,2)
9	Find the maximum sum of any two distinct nested tuples.	Input: T=((1,2),(5,1),(3,4))	Output: 9
10	Convert a nested tuple matrix into a dictionary using row-column coordinates.	Input: T=((10,20),(30,40))	Output: {(0,0):10,(0,1):20,(1,0):30,(1,1):40}
11	Find the tuple with the greatest number of distinct elements.	Input: T=[(1,1,2),(2,3,4),(5,5,5)]	Output: (2,3,4)
12	Find all index pairs whose tuple values are equal.	Input: T=(4,7,4,9,7)	Output: (0,2), (1,4)